Random Variables

Code authors: Jinhui Li, Viviana Monje.

Corresponding author: vmonje@buffalo.edu

Recommended Matlab: >= 2023a.

Table of Contents

Usage Rights....1
Learning Objectives....1
Practice Activities for Notebook 2....1
Activity 1: True or False....1
Activity 2: Number of Calls at a Customer Service Center....2
Activity 3: Delivery Time for Online Orders....2
Activity 4: Portfolio Return and Risk....3
Activity 5: Chocolate Bar Weights....3

Usage Rights

These materials were created for educational purposes and to be completed as part of material covered in the CE305: Probability, Statistics, and Data Analysis undergraduate course. Students will solve this practice notebook as part of the weekly assignment and turn it to be graded for completion and not correctness towards the participation category of their final grade.

Learning Objectives

By the end of this notebook, students will be able to:

- **Part I**:
- Define a random variable for an experiment;
- Differentiate between discrete and continuous random variables;
- Compute and interpret descriptive statistics of random variables;
- Differentiate between probability mass functions and probability density functions;
- Construct the appropriate probability distribution for a discrete and a continuous random variable;
- Use the definition of cumulative distribution functions (CDFs) to compute event probabilities;
- **Part II**
- Understand linear functions and combinations of random variables;
- Interpret the impact of sample size on the variability of statistical measures.
- Determine whether a continuous random variable is appropriate to describe data trends for a discrete random variable.

Practice Activities for Notebook 2

Submit this notebook along with your problem set assigned for Week 4 (due September 19th, 2025)

Activity 1: True or False

1. An online platform records each student's test score out of 100. The score is a discrete random variable.

2. A machine measures the exact length (in centimeters) of metal rods it cuts. The measured lengths can take infinitely many decimal values, and the length is modeled as a continuous random variable.

3. At a grocery store, the number of customers arriving every hour is a continuous random variable because the store is open all day.

4. An engineer tests how long a new light bulb lasts before it burns out. The lifetime is measured in hours and includes decimals, and it is a continuous random variable.

5. A mobile game tracks the number of times a player completes a level each day. This number is a continuous variable.

Activity 2: Number of Calls at a Customer Service Center

A customer service center receives a random number of phone calls per hour. Based on past records, the maximum number of calls in one hour is 3.

Questions:

- Define the experiment
- Define the ramdom variable
- Compute the cumulative distribution function F(x) for all $x \in \{0,1,2,3\}$ given the respective probabilities are: 0.15, 0.35, 0.30, 0.20.
- Find the expected number of calls E(X).
- Compute the standard deviation of X.
- What is the probability that the center receives **at least** 2 calls in one hour?

Activity 3: Delivery Time for Online Orders

A local courier company studied its delivery time (in hours) for same-day online orders. The data shows the delivery time, X, has the following continuous probability density function:

$$f(x) = \begin{cases} \frac{3}{8}x^2, & 0 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

Questions:

- Find the cumulative distribution function F(x).
- What is the probability that a package is delivered: a) within 1 hour; b) between 1 and 1.5 hours?
- Find the expected delivery time E(X).
- Find the variance Var(X) and standard deviation σ_X.

Activity 4: Portfolio Return and Risk

A small investor holds a portfolio with two types of stocks:

- Stock A has a mean return of 5% and standard deviation of 2%.
- Stock B has a mean return of 8% and standard deviation of 4%.

The investor puts 60% of their money in Stock A and 40% in Stock B. Let X_1 and X_2 represent the returns of Stock A and B, respectively. Assume stocks A and B are independent of each other.

Questions:

- What is the expected value, or expectation, of the total return for this combination?
- Compute the standard deviation of the total return.

Activity 5: Chocolate Bar Weights

A chocolate factory produces standard bars that should weigh exactly 100 grams. In reality, due to small variations in the machine, the bar weights follow a distribution with:

- Mean weight $\mu = 100$ grams
- Standard deviation $\sigma = 2$ grams

A quality control inspector randomly selects a sample of 25 bars to check if the average weight meets the standard. Let $\overline{X}$ be the sample mean of the 25 bars.

Questions:

- What is the expected value $\mu_{\overline{X}}$ of the sample mean?

- What is the standard deviation $\sigma_{\overline{X}}$ of the sample mean?

- If the inspector wants the standard deviation of the sample mean to be no more than 0.1 grams, what is the minimum sample size needed?